

ASSIGNMENT

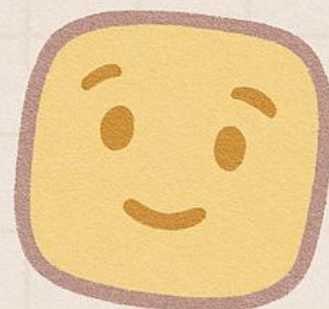


Name: Humna Imran

Assignment Topic: Basis Theorem

Teacher Name: Muhammad Jawad

Department: Artificial Intelligence
(Section A)



BASIS THEOREM



LINEAR ALGEBRA



BASISTHEOREM

INTRODUCTION

- Linear Algebra is important in real-world applications like Artificial Intelligence, Computer Graphics, and Data Science.
- The Basis Theorem tells us that a set of independent vectors can represent the entire vector space. 
- If vectors are independent and span the space, they form Basis

CONCLUSION

✓ Yes! $((1,0), (0,1))$ forms a basis for $\mathbb{R}^2$!



GOOD JOB!

IMPORTANCE OF BASIS THEOREM

- Simplifies complex systems
- Essential for Machine Learning and Data transformations.
- Helps in building better simulations 

EXAMPLE PROBLEM

Q1: Check if $(1,0), (0,1)$ is a basis for $\mathbb{R}^2$

- Step 1. Check Linear Independence
- Suppose $a(1,0)+b(0,1) = (0,0)$ Expanding gives $(a,b) = (0,0)$
- Therefore $a=0, b=0$ ✓
- Linearly Independent!
- They span $\mathbb{R}^2$.



WORKED EXAMPLES: BASIS CHECK

Q2: Determine if $((2,4), (1,2))$ forms a basis for $\mathbb{R}^2$

✓ **Step 1:** Check Linear Independence

↪ Set up: $a(2,4) + b(1,2) = (0,0)$

Expand: $(2a+b), 4a+2b) = 0$

↪ System of equations:

$$2a + b = 0 \quad | \quad 4a + 2b = 0$$

↪ **Step 2:** Solve the System

✓ Infinitely many solutions

✗ → Linearly Dependent! →  (True)

No, $((2,4), (1,2))$ do not form a basis for $\mathbb{R}^2$.

Q3: Check if $(1,2,3), (4,5,6), (7,8,9))$ forms a basis for $\mathbb{R}^3$

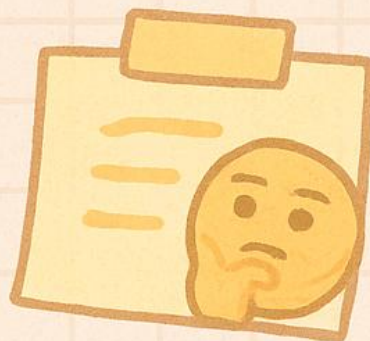
✓ **Step 1:** Check Linear Independence

↪ Subtract $a(1,2,3) + b(4,5,6) + c(7,8,9) = (0,0,0)$
 $= a+b+c=0 \quad | \quad (a+b+c=0 \text{ (Eq 4)})$

↪ Subtract Equation 2 from Equation 3:
 $(3a+6b+9c=0 \quad | \quad a+b+c=0 \text{ (Eq 5)})$

No, $((1,2,3), (4,5,6))$ do not form a basis for $\mathbb{R}^3$.

WORKED EXAMPLES: BASIS CHECK



Q4: Determine if $\{(1,2,1), (3,4,0), (2,6,3)\}$ forms a basis for $\mathbb{R}^3$.

➤ Step-by-Step Solution

➤ Step 1: Check Linear Independence



➤ From $(a+3c-0) \rightarrow -3c$

Substitute into first eq: $3b \rightarrow bc=0$

➤ $b+c \Rightarrow 3bc+bc=0 \rightarrow b=c$

➤ Thus, $a=b=c=0$.

➤ Step 3: Span Check $\rightarrow$ they span $\mathbb{R}^3$.

Q5: Determine if $\{(1,1,1,0), (1,1,0,1), (0,1,1,1), (1,0,0,1)\}$ forms a basis for $\mathbb{R}^4$.

➤ Step 1: Check Linear Independence



➤ $t(1,1,1,0) + b(1,1,0,1) + c(0,1,1,1) + d(1,0,0,1)$

Substitute into Equ, 3 $\rightarrow c \Rightarrow$

➤ $a=b \neq b+c=0 \rightarrow c=d$

➤ $b=c=c=d=0$

➤ Step 3: Solve the $t \rightarrow$ they span $\mathbb{R}^4$.

$$a+b+c=0$$

$$a+c+d=0$$

$$b+c=d=0$$